

connected to the bases of transistors T1 through T16 via resistors R6 through

R21, respectively. All the relays (RL1 through RL16) are connected to

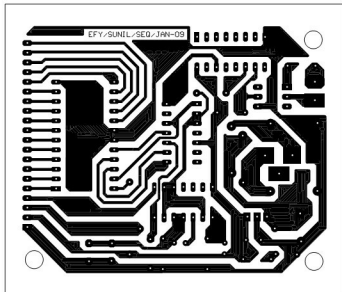


Fig. 4: Actual-size, single-side PCB for the logic control section of sequential power-on'

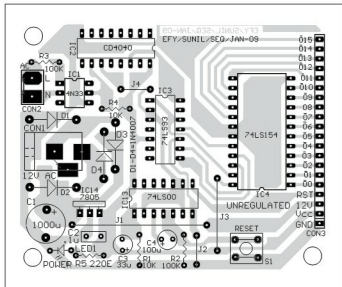


Fig. 5: Component layout for the PCB shown in Fig. 4

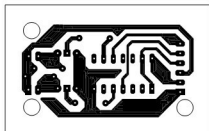


Fig. 6: Actual-size, single-side PCB for the relay driver section of sequential power-on'

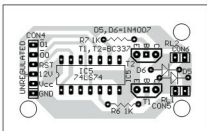


Fig. 7: Component layout for the PCB in Fig. 5

the collectors of transistors T1 through T16, respectively. Diodes D5 through D20 connected across relays RL1 through RL16, respectively, act as free-wheeling diodes.

When output pin 5 of flip-flop 74LS74 goes high, transistor T1 is driven into saturation and relay RL1 energises. Similarly, the high 'Q' outputs of other flip-flops drive relays RL2 through RL16.

Flip-flops can energise relays RL1 through RL16 randomly when mains power resumes. To avoid the random energisation of relays, power-on reset is achieved

with NAND gate 74LS00. The NAND gates are configured as a monostable. Reset pin R of IC2 and IC3 is connected to pin 6 of NAND gate N2. The output of N2 is inverted and connected to reset R input of all the flip-flops. Switch S1 is used for manual reset.

The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC14), which provides +5V DC output. Capacitor C2 does further filtering. LED1 indicates DC power-on, while resistor R5 acts as a current limiter.

Construction and testing

Actual-size, single-side PCBs for the logic control section (on the left side of the dotted line) and relay driver section (on the right side of the dotted line) of the sequential power-on' circuit are shown in Figs 4 and 6, and their component layouts in Figs 5 and 7, respectively. Connectors have been provided on the PCB for all the outputs (O0 through O15) of IC4, RST, unregulated 12V, Vcc and GND. Thus it can be extended for the relay driver section at any desired location. The PCB of the relay driver is made for two relays to drive two loads. You can extend the same to more than two loads by adding another PCB as shown in Fig. 6. Wire the circuit on the PCB shown in Fig. 6 and extend it to any other equipment in the house. Use bases for all the ICs. Check continuity between respective connections using a multimeter. Connect the electrical loads through relay contacts as shown in Fig. 3.

One after another load is switched 'on' every 20 seconds. Manually reset all the loads if any of the sixteen loads turns on after power resumption. Use mains-rated cable for all load connections. ●